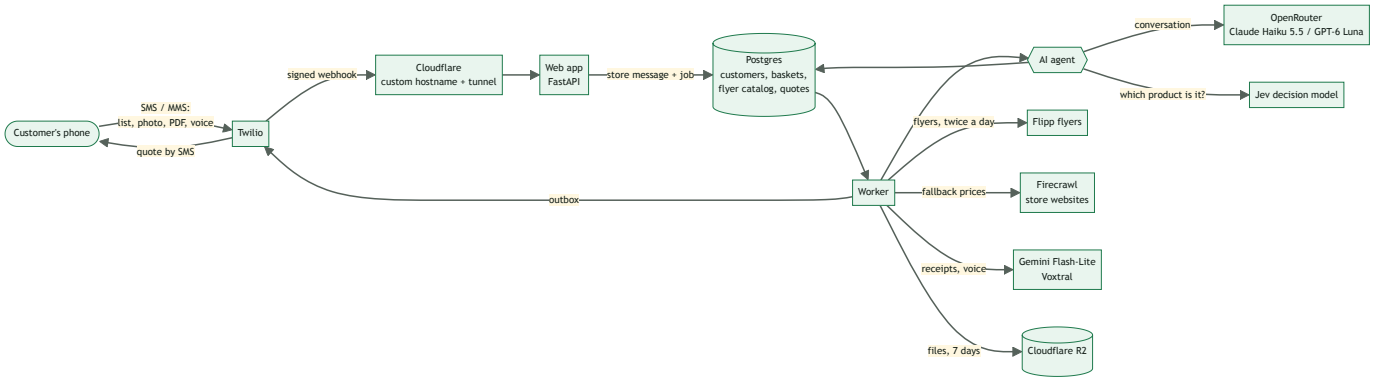
Dollar Hunter

Architecture and logic, end to end. A grocery-savings assistant that works entirely by text message.

1. The system at a glance

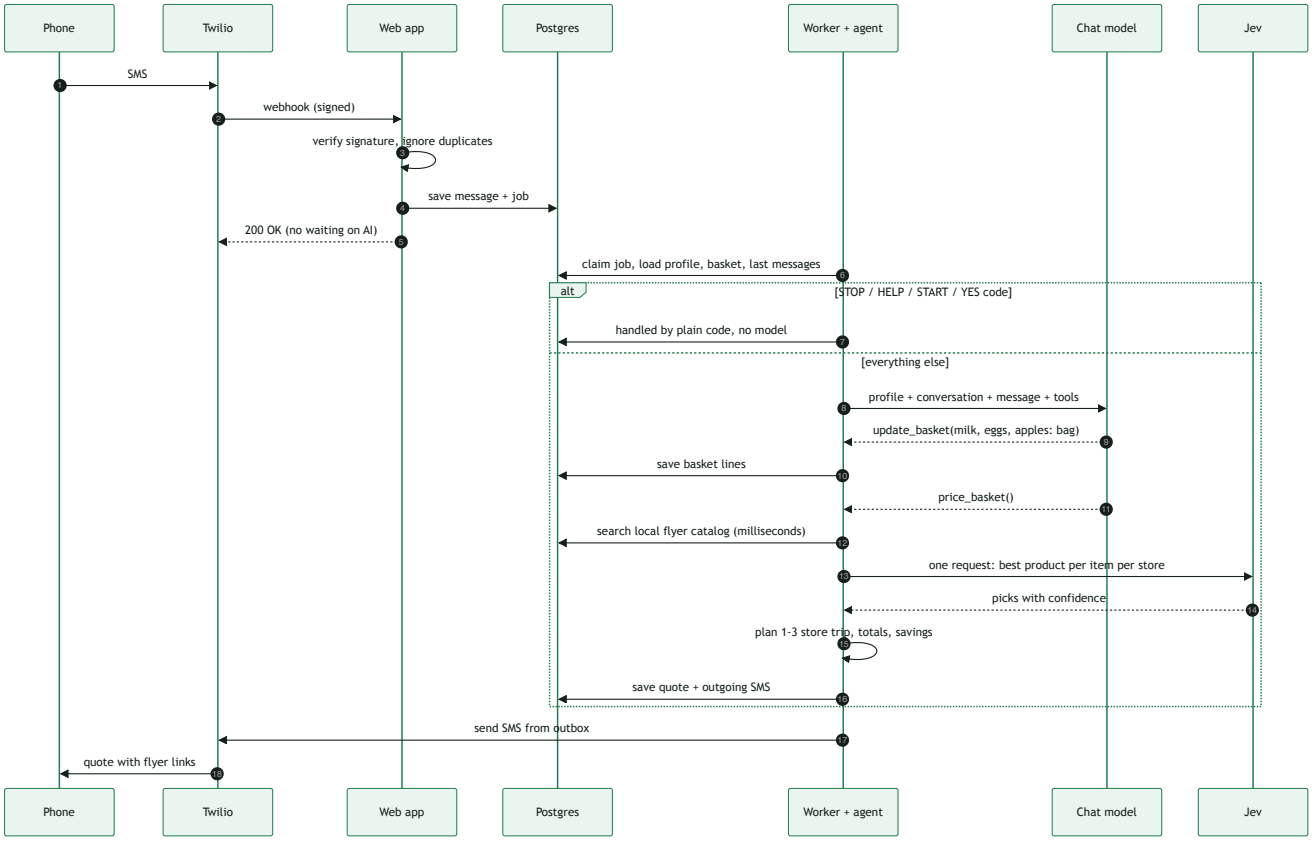
A customer texts a phone number. Everything else happens on one small server: the message is stored, an AI agent works out what they mean, code looks up this week's flyer prices near them, and a quote goes back by SMS.



Part	Job
Web app	Receives the Twilio webhook, checks its signature, saves the message and a job, and answers Twilio in milliseconds. Also serves the landing page.
Worker	Runs the jobs: the agent, receipt and voice processing, sending texts, flyer downloads, clean-up.
Postgres	The memory: one customer per phone number, with their location, preferences, baskets, quotes, messages, plus the shared flyer catalog.
Models	The conversation model understands people and calls tools. Jev only picks matching products. Neither ever writes a price.

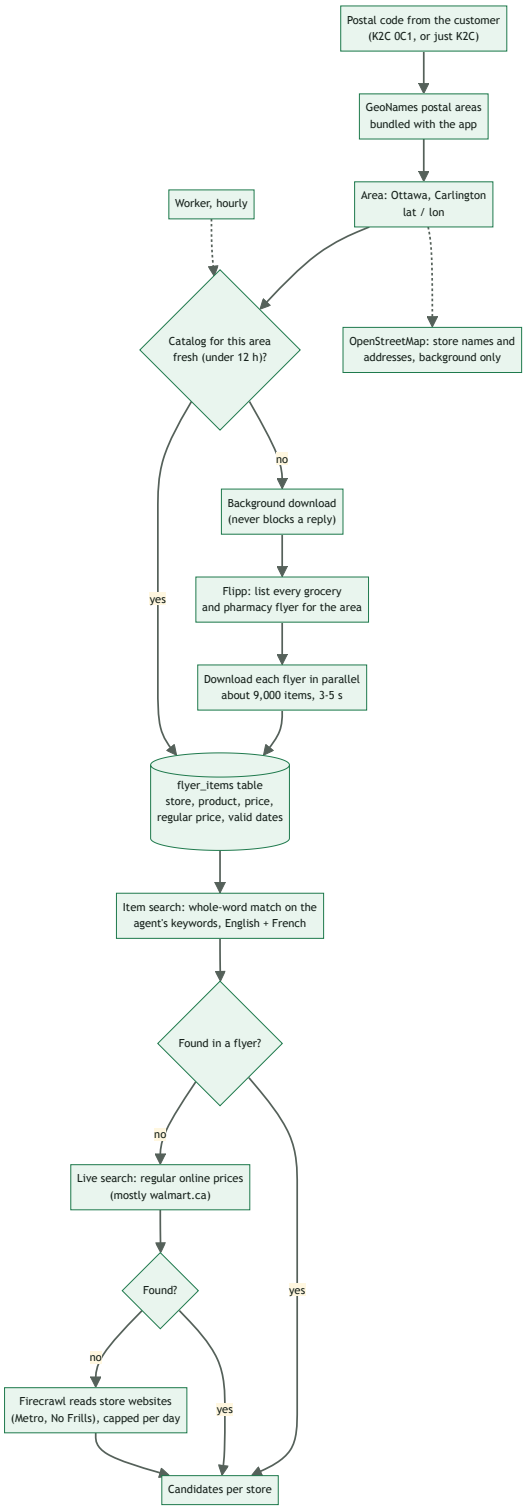
2. One message, end to end

What happens between "need milk, eggs and a bag of apples, what's the total?" and the quote arriving (typically 2 to 6 seconds).



3. Where the data comes from

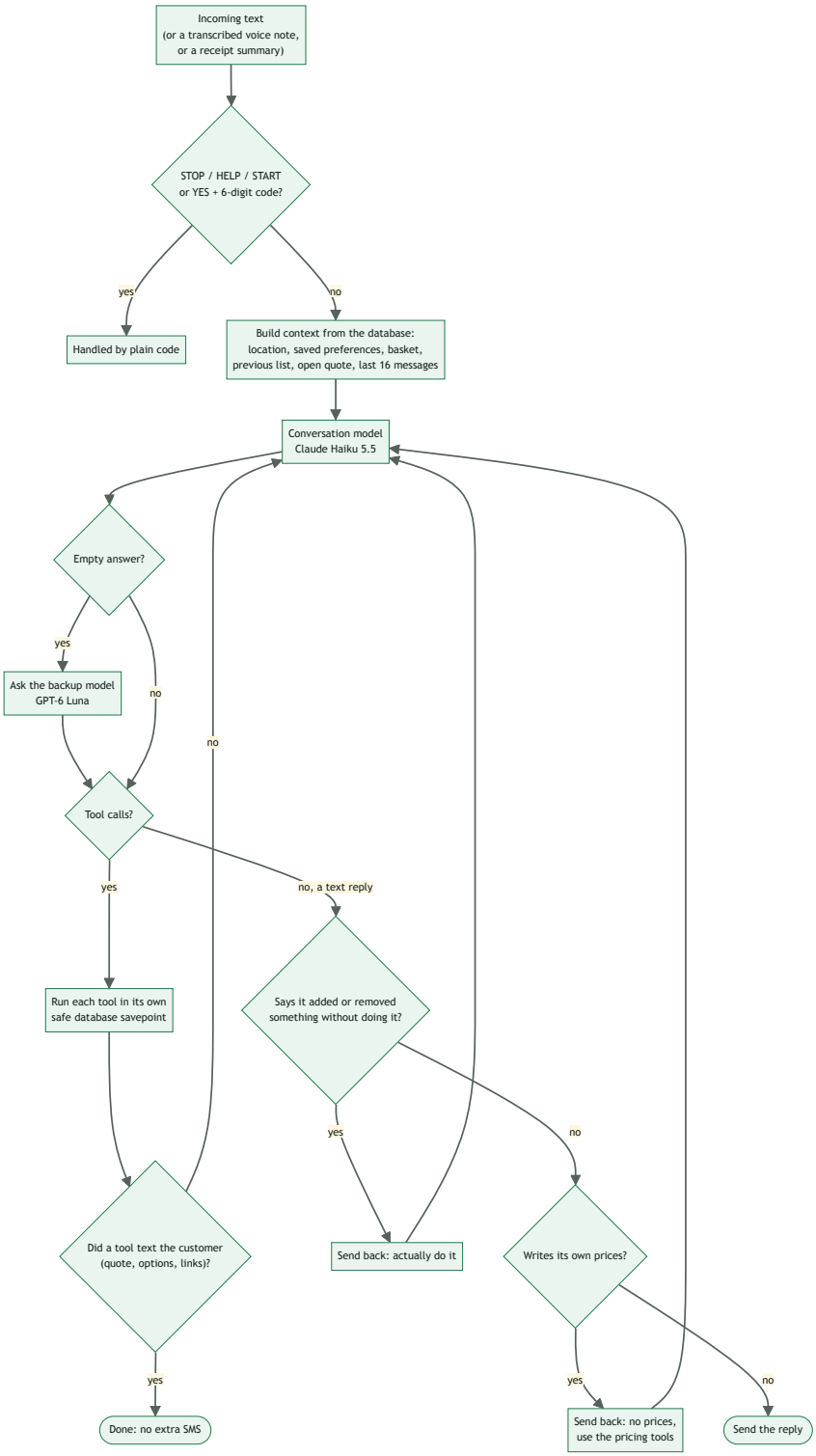
Prices are gathered ahead of time, so answering is a database lookup, not a slow web search.



- **Regular price** comes from the flyer's own discount (a 36% off item gives its regular price), so savings are checkable. Items without it carry no saving and no fee.
- **Caching:** flyer catalog 12 h per area, live searches 6 h, website prices 3 days, store lists 7 days, Jev decisions 6 h (the same basket always gets the same picks).
- **Honest limits:** Flipp has no official API; this uses the endpoints its own app uses. Store websites show their default branch.

4. How the agent thinks

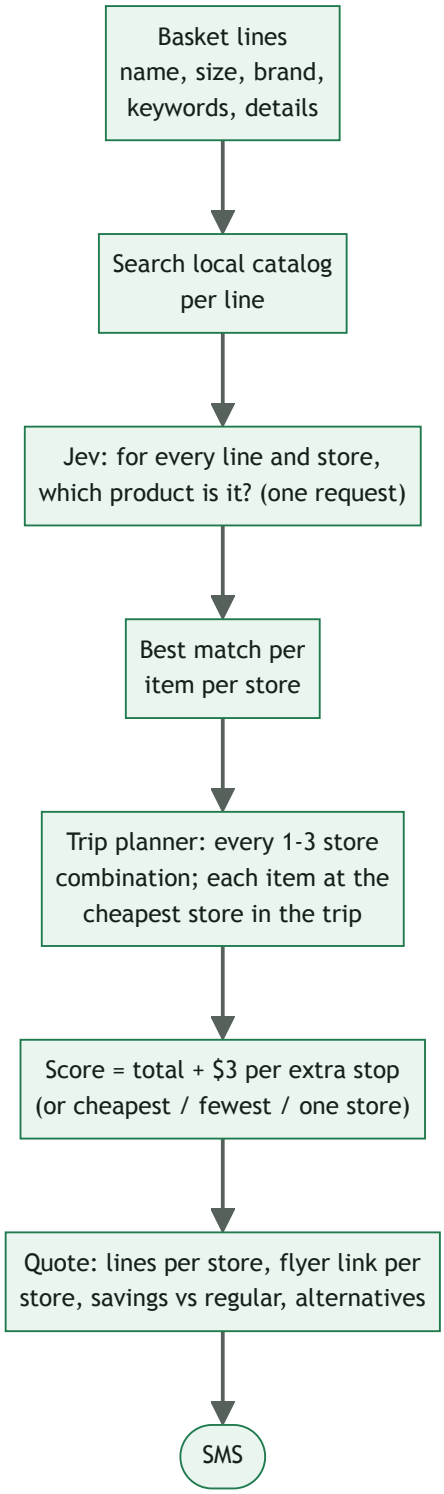
The conversation model decides what to do; code does everything involving money, and checks the model's work.



Tool	What it does
set_location, find_stores	Save the postal area; name the chains with flyers there.
update_basket	Add, change or remove items with quantity, size, brand, search keywords and a short description of what this customer means.

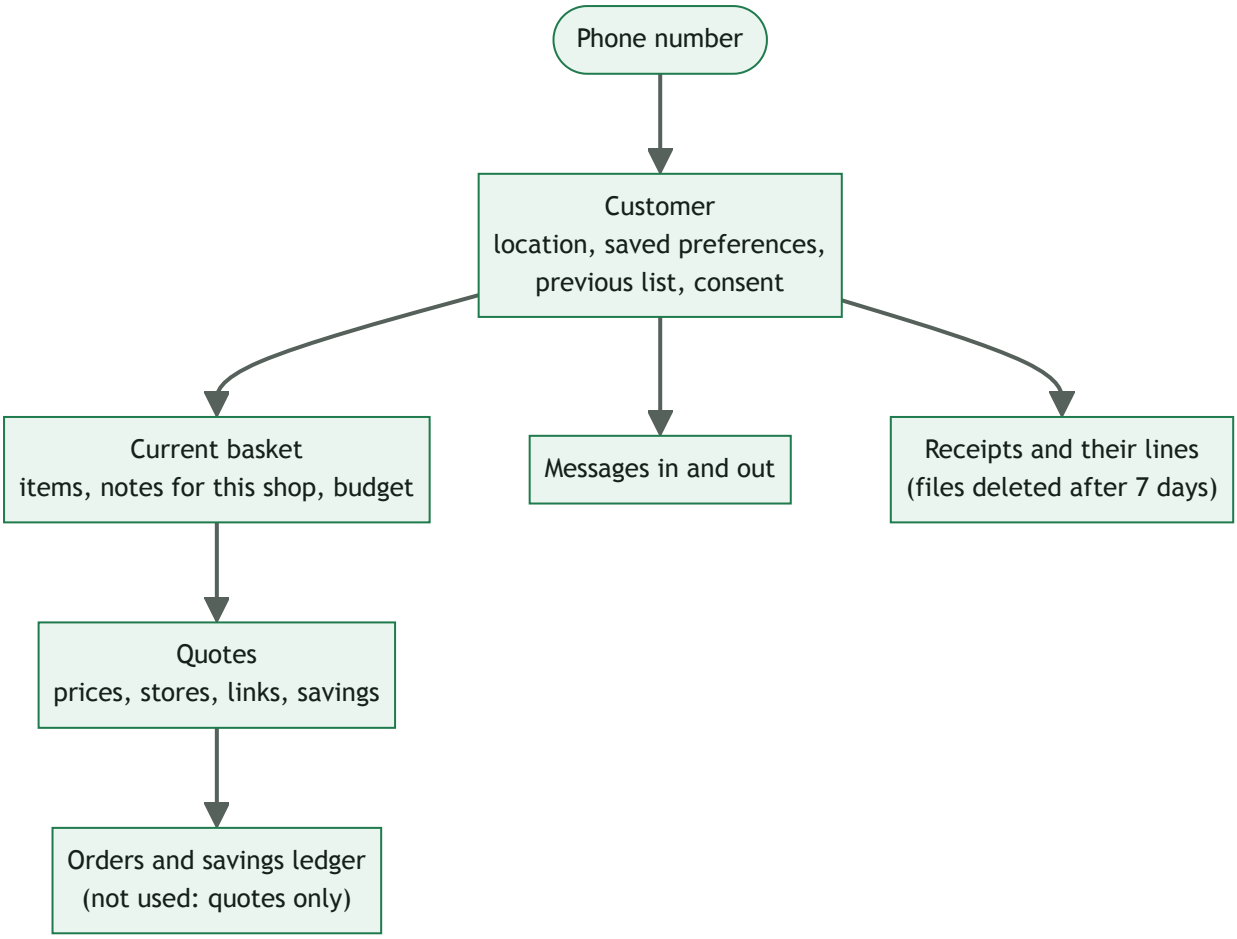
show_options, choose_option	Text the cheapest real options for one item; pin the one they pick.
price_basket	Plan the trip and text the quote (balanced, cheapest anywhere, fewest stops, or one named store).
send_links	Direct flyer link for each item in the quote.
remember, forget, clear_basket, copy_last_order, set_budget, accept_receipt_items	Memory and list management.

5. From basket to quote



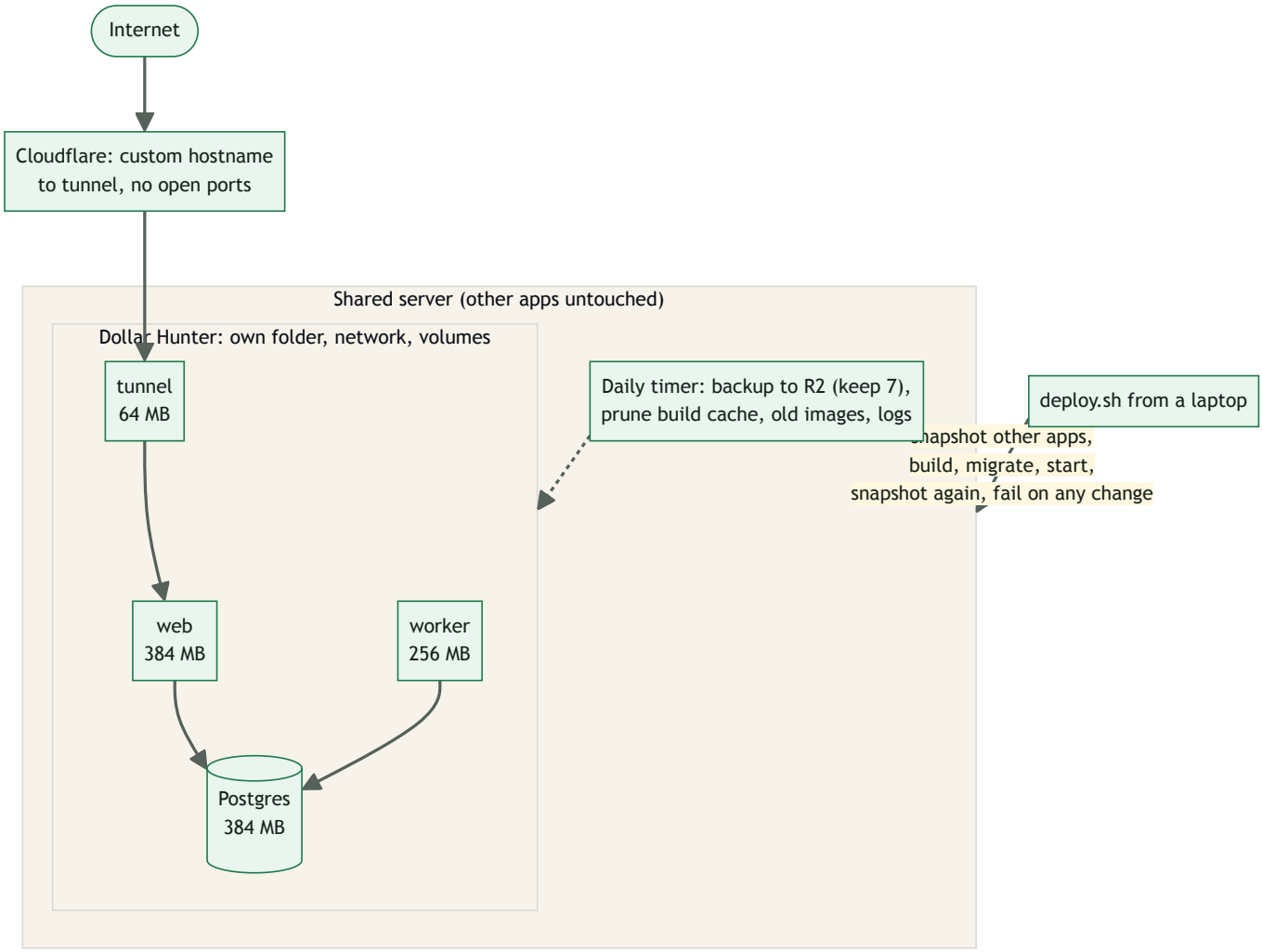
- **Savings** = regular price minus flyer price, only where the flyer states the regular price. **Fee** = 25% of verified savings; none without them.
- **Alternatives shown with every quote:** buying everything at one store, the absolute cheapest spread, or the fewest stops, so the customer sees the trade-off.
- Texts are kept in the GSM-7 alphabet (symbols such as ® removed, bilingual names reduced to English), which keeps the number of billed SMS segments down.

6. Memory per phone number



"Clear my list" empties only the basket. The old list stays as *previous list*, so the agent can say what was on it and add items back on request; preferences, location and history stay.

7. Running it: isolation and upkeep



- **No ports published:** traffic arrives only through the Dollar Hunter Cloudflare tunnel; nothing else on the host is touched.
- **Self-cleaning:** logs capped at 10 MB per container, receipts deleted after 7 days, finished jobs after 14 days, messages after 90 days, previous and current image kept for rollback.
- **Removal:** one script deletes its containers, volumes, images, build cache, timer and folder.
- **Costs per message:** chat model about $0.002, Jev about $0.0004, a receipt under $0.001, a quote 3 to 5 SMS segments.